

Nuthan Prasad S

7795308852 | nuthanpdd@gmail.com | [linkedin](#) | [github](#)

Summary

AI Engineer with 4+ years of experience building scalable backend systems and enterprise-grade Agentic AI solutions. Proven expertise in designing multi-agent workflows, RAG pipelines, and LLM-powered applications using Python, LangChain, and modern AI frameworks.

Education

Bengaluru Central University

Bachelor of Computer Science and Application

2018 – 2021

Bengaluru, KA

AI Engineering Projects

Enterprise RAG Assistant

- Built enterprise chatbot using OpenAI, LangChain, and Vector Databases for document-based Q&A; and internal knowledge search using RAG architecture.
- Created application for displaying LLM-generated product review summaries, sentiment insights, and recommendation analysis.

Work Experience

Tata Consultancy Services (TCS)

System Engineer

Jul 2025 – Present

Bengaluru, KA

- Developed advanced **RAG pipelines** integrating vector databases (ChromaDB, FAISS) with enterprise document sources, improving contextual response accuracy by 35% while maintaining sub-X ms retrieval latency.
- Implemented observability and evaluation pipelines using **LangSmith**, **CloudWatch**, and custom telemetry dashboards to monitor **LLM latency**, **token consumption**, **hallucination rates**, and agent reasoning traces across X+ daily interactions.
- Architected and deployed enterprise-grade Agentic AI applications using **LangChain**, **LangGraph**, and **Google ADK**, enabling intelligent multi-step reasoning workflows and reducing manual operational effort by 45%.
- Integrated enterprise APIs and databases, enabling seamless interaction between AI agents and HR systems.

Capgemini Technology Services India Limited

Software Engineer

Dec 2021 – Jul 2025

Bengaluru, KA

- Integrated AI systems with enterprise platforms, **REST APIs**, **PostgreSQL databases**, and cloud-native services (**AWS Lambda**, **EC2**, **S3**), enabling scalable deployment and seamless ingestion of structured and unstructured data sources.
- Automated** CI/CD and deployment workflows for AI services using Docker, Jenkins, and cloud infrastructure, reducing deployment time by 40% and improving production stability for AI-driven applications.
- Debugged REST API integration failures across microservices including HTTP timeouts, OAuth token flows, and payload serialisation errors.

Technical Skills

AI & Agentic Frameworks: LangChain, LangGraph, LLM APIs (OpenAI, Gemini), RAG Pipelines, Prompt Engineering.

Vector Databases & Embeddings: ChromaDB, FAISS, OpenAI Embeddings, HuggingFace Embeddings, Semantic Search, Chunking Strategies, Metadata Filtering, Vector Search, Hybrid Search, Vertex AI, Tool Calling, Function Calling.

Languages & Frameworks: Python (FastAPI, Pydantic), RESTful APIs, SQL (PostgreSQL, MySQL), Bash/Shell

Cloud & DevOps: AWS (EC2, S3, VPC, CloudWatch, Lambda, RDS, IAM), Docker, Jenkins, CI/CD Pipelines, Git, GitHub

Observability & Monitoring: LangSmith, New Relic, Splunk, AWS CloudWatch.

Developer Tools: Postman, JIRA, Confluence, GitHub, Bitbucket, Linux, IntelliJ, VS Code, Agile / Scrum

Certifications

- AWS Certified Cloud Practitioner — Amazon Web Services
- Prompt Design in Vertex AI — Google Cloud